



Field control of an electromagnet in mT.

cIMag drives the magnet current from a bipolar supply and closes the loop on a Hall probe, using a measured calibration of the magnet including its hysteresis. It runs in simulation; it has not yet been tested on the magnet, and the control gains are simulator values that need tuning on the real magnet. The window has a dark and a light theme.

Kepeco BOP bipolar supply · GMW 3470 electromagnet · Hall probe on an NI USB-6259 DAQ

Module source on GitHub

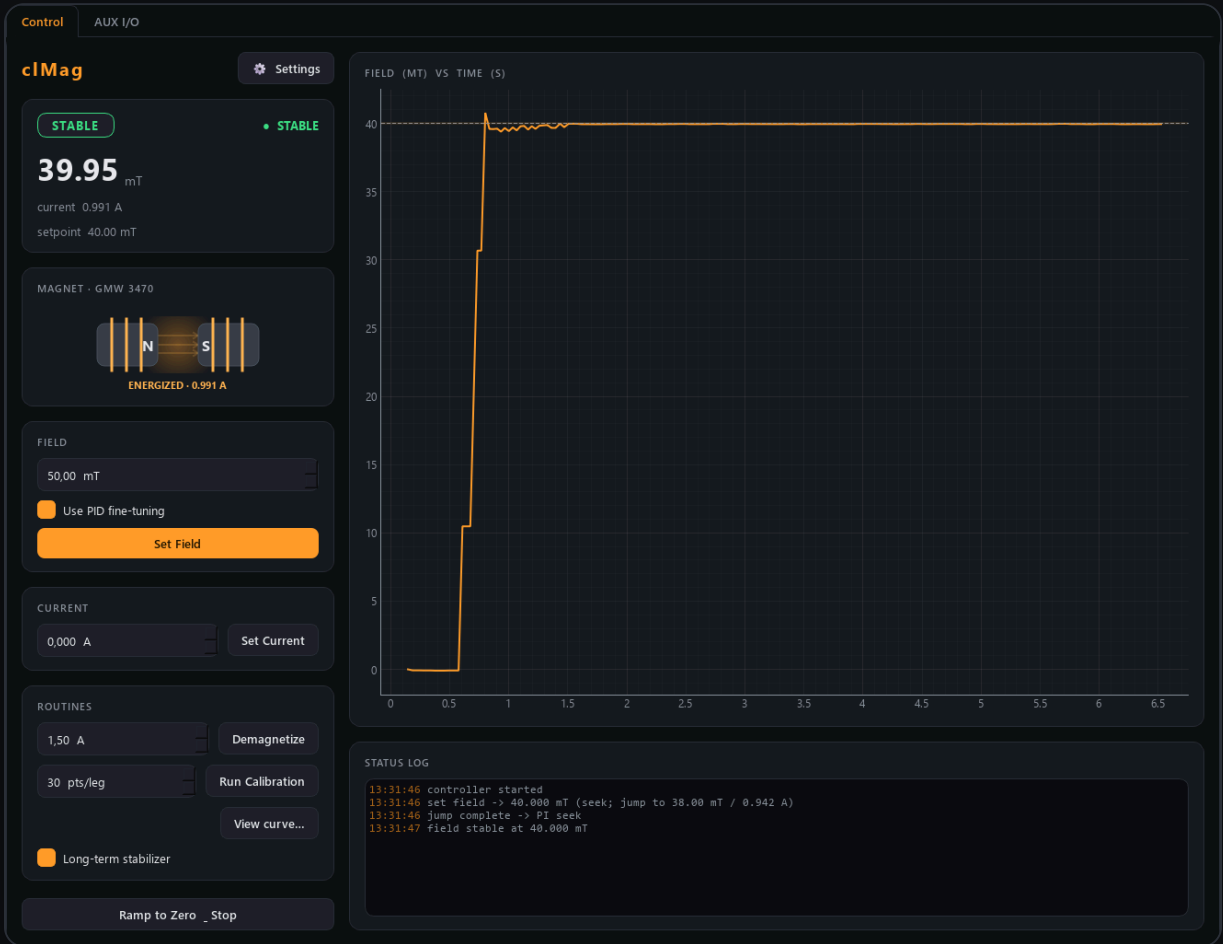
AaltoFlow on GitHub

field calibration PI with output freeze drift stabiliser demagnetise simulator included

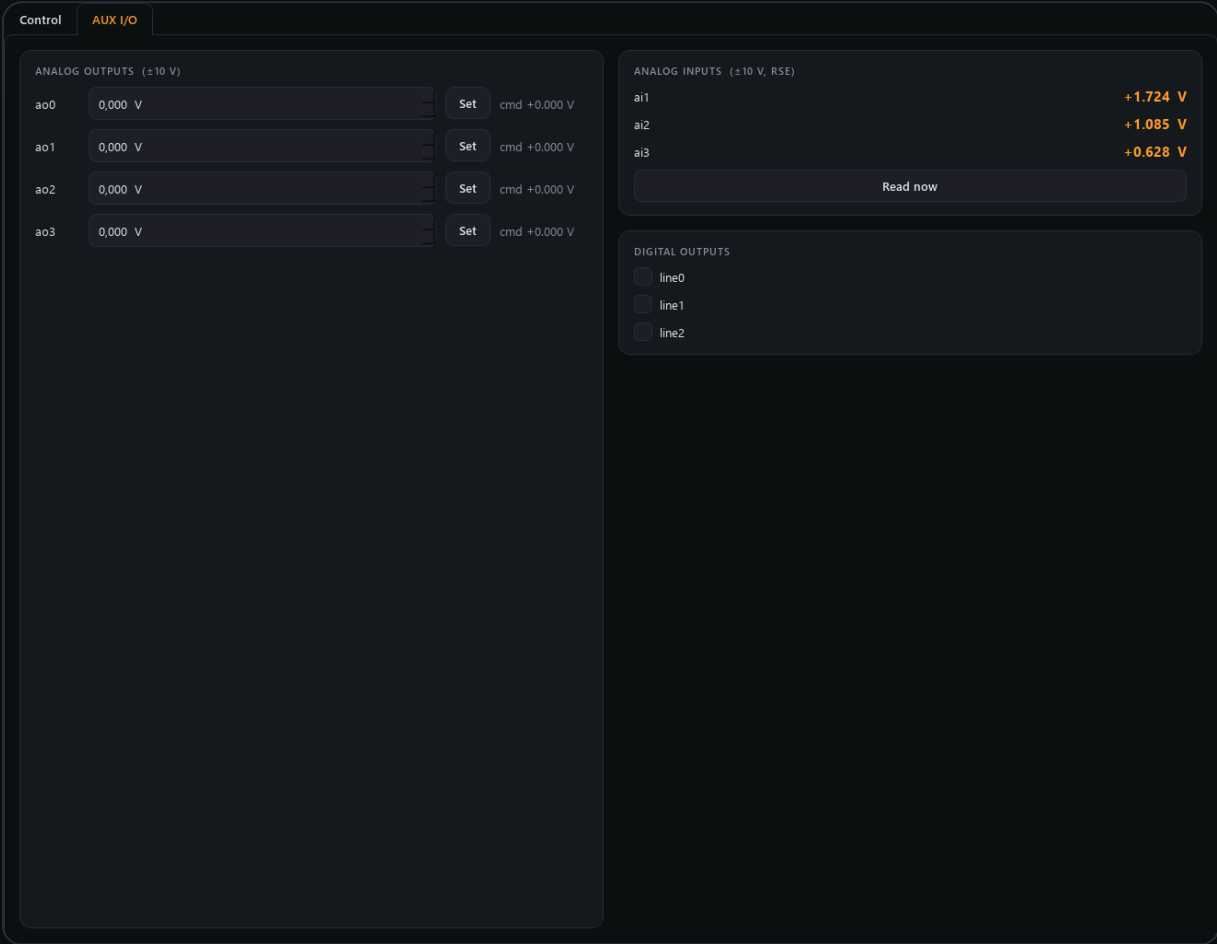
CONTROL

Calibration first, PI only for the last step

The measured B(I) calibration, with its up and down branches, is used to jump the current straight to a value just below the target field, so the field gets close quickly without overshoot. Only then does a fast PI loop (field read every ~4 ms) correct the remaining difference, approaching from one side. Once the field is within half the tolerance the output is frozen; without the freeze, hysteresis makes a PI loop oscillate around the target.



Control tab: field against time, setpoint and stability indicator (simulation)



AUX I/O tab

EXTRA I/O

Other channels of the DAQ

Four analog outputs, three analog inputs and three digital lines of the same DAQ, available from the window, the console and scans.

Calibration

Measures B(I) on the up and down branches; view, save and load it. The field range offered to scans follows the calibration.

Stabiliser

In hold, a slow correction with a dead band compensates drift.

Demagnetise

Decaying alternating steps to zero field.

Safety

Current limit; ramp to zero and output off on shutdown, on a crash or when the controlling client disconnects.

Scans

A scan sets the field and waits until the module reports the new field as stable.

Network

The service can run on the lab PC and its window on another PC.

PART OF AALTOFLOW

AaltoFlow

In AaltoFlow each instrument runs as its own service with its own window. A launcher starts them, and a scan engine drives any combination of them through multi-dimensional measurements.

[github.com/FlashLukas/AaltoFlow](#)
[cIMag-control on GitHub](#)

LICENCE AND CITATION

Citation

Released under the MIT licence. If you publish data measured with it, please cite:

L. Flajšman, *AaltoFlow: lab automation built from independent instrument modules*, NanoSpin group, Aalto University, doi:10.5281/zenodo.22959230

Copy citation